FINITE GROUPS WITH TWO CONJUGACY CLASSES OF p-ELEMENTS AND RELATED QUESTIONS FOR p-BLOCKS

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ABSTRACT. We prove that a finite group in which any two nontrivial p-elements are conjugate have Sylow p-subgroups which are either elementary abelian or extraspecial of order p^3 and exponent p.

1. INTRODUCTION

Let G be a finite group and let p be a prime. The main result of this note is to solve a group theoretical problem that naturally arises in block theory (and in the theory of fusion systems).

Theorem A. *Let p be a prime and G a finite group in which any two nontrivial p-elements are conjugate. Then one of the following holds:*

(i) *The Sylow p-subgroups of G are elementary abelian;*
(ii) *$p = 3$ and $\mathbf{O}^{p'}(G/\mathbf{O}_{p'}(G))$ is isomorphic to Ru, J_4 or ${}^2F_4(q)'$ with $q = 2^{6b\pm1}$ for a nonnegative integer b;*
(iii) *$p = 5$ and $G/\mathbf{O}_{p'}(G)$ is isomorphic to Th.*

In cases (ii) and (iii), the Sylow p-subgroups of G are extra-special of order p^3 and exponent p.

Theorem A can be reformulated in terms of Brauer blocks. If B is a p-block of a finite group G, recall that a *B-subsection* is a pair (u, b_u) consisting of a p-element $u \in G$ and a p-block b_u of the centralizer $\mathbf{C}_G(u)$ such that the induced block $(b_u)^G = B$. The *trivial* B-subsection is $(1, B)$. It is a natural problem to study blocks in which all the non-trivial subsections are G-conjugate, and the question is whether or not in this case the defect groups of B should be elementary abelian or extra-special of order p^3 and exponent p. Theorem A provides an answer to this for the principal block of G. (In Section 3 of this paper, we shall comment on the corresponding question for fusion systems.)

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Theorem A is also related to a recent theorem due to the second and fourth authors of this paper, which states that for $p \neq 3, 5$, a finite group G has abelian Sylow p-subgroups if and only if the conjugacy class size of every p-element of G is not divisible by p, cf. [23]. This group theoretical result also admits a block theoretical reformulation which we find of interest. We discuss this and some other related problems in Section 3 of this paper. Finally, in Section 4, we study blocks with a small number of irreducible characters, a topic that naturally connects with the previous parts of the paper.

2. Proof of Theorem A

We begin by proving a slightly more general result than Theorem A.

Theorem 2.1. *Let p be a prime and G a finite group in which any two nontrivial cyclic p-subgroups are conjugate. Then one of the following holds:*

(i) *The Sylow p-subgroups of G are elementary abelian;*
(ii) *$p = 3$ and $\mathbf{O}^{p'}(G/\mathbf{O}_{p'}(G))$ is isomorphic to Ru, J_4 or ${}^2F_4(q)'$ with $q = 2^{6b\pm1}$ for a nonnegative integer b;*
(iii) *$p = 5$ and $G/\mathbf{O}_{p'}(G)$ is isomorphic to Th.*

As might be expected, the proof of Theorem 2.1 uses the Classification of Finite Simple Groups.

Note that Theorem 2.1 is trivial in the case where p does not divide $|G|$. So in proving Theorem 2.1 we will assume that $|G|$ is divisible by p. Then the assumptions in Theorem 2.1 imply that the exponent of P, $\exp(P)$, is p if $P \in \mathrm{Syl}_p(G)$. In particular, if $p = 2$, then certainly P is elementary abelian. Furthermore we have that $|g^G|$ has the same size for all $1 \neq g \in P$. By choosing $g \in \mathbf{Z}(P)$, we have that this common size is coprime to p. Hence, if $p > 5$ then Theorem 2.1 follows from the main result of [23]. So in what follows we may assume that $p = 3$ or 5.

Next we prove Theorem 2.1 for finite simple groups.

Theorem 2.2. *Let S be a finite simple group and let $p \in \{3, 5\}$ be a prime divisor of $|S|$. Suppose that $|g^S|$ is the same (and hence coprime to p) for all the nontrivial p-elements $g \in S$, and moreover assume that $\exp(P) = p$ for $P \in \mathrm{Syl}_p(S)$. Then one of the following holds:*

(i) *The Sylow p-subgroups of S are elementary abelian.*
(ii) *$P \cong p_+^{1+2}$ is extra-special of order p^3. Furthermore, either $p = 3$ and S is isomorphic to Ru, J_4 or ${}^2F_4(q)'$ with $q = 2^{6b\pm1}$ for a nonnegative integer b, or $p = 5$ and $S \cong Th$.*

Lemma 2.3. *Theorem 2.2 holds if S is an alternating or sporadic simple group.*

Proof. Repeat the proof of [23, Lemma 2.2]. □

Lemma 2.4. *Theorem 2.2 holds if S is a finite simple group of Lie type in characteristic p.*

Proof. It is convenient to view S as $[H, H]$, where $H = \mathcal{G}^F$ for a simple algebraic group $\mathcal{G}$ of adjoint type and a Frobenius map $F : \mathcal{G} \to \mathcal{G}$; in particular, $\mathbf{Z}(\mathcal{G}) = 1$ and $|H : S|$ is coprime to p. Suppose first that p is a good prime for $\mathcal{G}$. Then the proof of [23, Lemma 2.3] shows that $\mathcal{G}$ is of type A_1, i.e. $S = PSL_2(q)$ and so its Sylow p-subgroups are abelian.

Assume now that p is a bad prime for $\mathcal{G}$; in particular, $\mathcal{G}$ is exceptional since $p > 2$. Suppose that $\mathcal{G}$ is of type G_2, and so $p = 3$. If $H = {}^2G_2(q)$, by [19, Table 22.2.7] we can find a p-element $u \in H$ such that $|\mathbf{C}_H(u)| = 2q^2$, whence $q||u^H|$. Similarly, if $H = G_2(q)$, by [19, Table 22.2.6] we can find a p-element $u \in H$ such that $|\mathbf{C}_H(u)| = 2q^4$, whence $q^2||u^H|$. If $H = F_4(q)$ (and $p = 3$), by [19, Table 22.2.4] we can find a p-element $u \in H$ such that $|\mathbf{C}_H(u)| = 2q^6$, whence $q^6||u^H|$. If $H = E_6(q)$ or ${}^2E_6(q)$, by [19, Table 22.2.3] we can find a p-element $u \in H$ such that $|\mathbf{C}_H(u)| = q^8$, whence $q^{28}||u^H|$. If $H = E_7(q)$, by [19, Table 22.2.2] we can find a p-element $u \in H$ such that $|\mathbf{C}_H(u)| = 2q^{21}$, whence $q^{42}||u^H|$. Finally, if $H = E_8(q)$, by [19, Table 22.2.1] we can find a p-element $u \in H$ such that $|\mathbf{C}_H(u)| = 2q^{28}$, whence $q^{92}||u^H|$. □

In what follows, we use the notation SL^ϵ to denote SL when $\epsilon = +$ or $+1$, and SU when $\epsilon = -$ or -1, and similarly for the Lie types A_n^ϵ. Furthermore, N_p denotes the p-part of the integer N. We use the notation for various finite classical groups as described in [16].

Lemma 2.5. *Theorem 2.2 holds if the simple group S is a finite classical group in characteristic $r \neq p$.*

Proof. (i) First we consider the case $S = PSL_n^\epsilon(q)$, where $q = r^f$ and $\epsilon = \pm 1$. Since $p \neq r$ divides $|S|$, there is a smallest positive integer $m \leq n$ such that $p|(q^m - \epsilon^m)$. View $S = L/\mathbf{Z}(L)$, with $L = SL_n^\epsilon(q)$. Now the proof of [23, Proposition 2.5] shows that $m = 1$ and $n = p = 3$. Suppose that $9|(q - \epsilon)$. Choosing $\theta \in \overline{\mathbf{F}}_q$ of order 9 and $g = \mathrm{diag}(\theta, \theta^{-1}, 1) \in L$, we see that g has order 9 in S, contradicting the condition that $\exp(P) = p$. On the other hand, if $9 \not| (q - \epsilon)$, then P is elementary abelian of order 9.

(ii) For all other classical groups, since $p > 2$ we may without any loss replace S by $G = Sp_{2n}(q)$, $GO_{2n+1}(q)$, $GO^+_{2n}(q)$, or $GO^-_{2n}(q)$, if S is of type C_n, B_n, D_n, or ${}^2D_n$, respectively. In all these cases, there is a smallest positive integer $m \leq n$ such that $p|(q^{2m} - 1)$, and $\alpha = \pm 1$ such that $p|(q^m - \alpha)$. Write $(q^m - \alpha)_p = p^c$. Since $p > 2$ we have that $(q^{2mp} - 1)_p = (q^{mp} - \alpha)_p = p^{c+1}$. If $n < mp$, then the proof of [23, Proposition 2.5] shows that the Sylow p-subgroups of S are abelian. So we may assume that $n \geq mp$.

Suppose first that $(q,m,p,\alpha)=(2,1,3,-)$. In this case $Sp_6(2)$ embeds in G and so P contains an element of order 9, a contradiction.

Hence we may assume that $(q,m,p,\alpha)\neq(2,1,3,-)$. Then the proof of [23, Proposition 2.5] shows that G contains an element of order $p^{c+1}>p$, unless $G=GO^{-\alpha}_{mp}(q)$, in which case the Sylow p-subgroups of S are abelian. □

Lemma 2.6. *Theorem 2.2 holds if S is an exceptional finite simple group of Lie type in characteristic $r\neq p$.*

Proof. (i) The case $S={}^2F_4(2)'$ can be checked directly using [4], so we will assume $S\ncong{}^2F_4(2)'$. We view $S=G/\mathbf{Z}(G)$, where $G=\mathcal{G}^F$ for a simple algebraic group $\mathcal{G}$ of simply connected type and a Frobenius map $F:\mathcal{G}\to\mathcal{G}$. If $G\in\{{}^2B_2(q),\ {}^2G_2(q)\}$ is a Suzuki or a Ree group, then the Sylow p-subgroups of G are all abelian as $p>2$. The same is true when $G=S={}^2F_4(q)$ with $q\geq 8$, unless $p=3$. In the latter case, $p=3|(q+1)$; moreover, according to [18, Table 5.1], G contains a subgroup $X\cong SU_3(q)$. If $9|(q+1)$, then, as we noted in the proof of Lemma 2.5, X contains an element of order 9, contrary to $\exp(P)=3$. Assume that $9\not|(q+1)$, which means that $q=2^{6b\pm1}$ for some $b\geq 1$. Then $|S|_3=27$, and P can be embedded in ${}^2F_4(2)'$, whence $P\cong 3^{1+2}_+$.

(ii) In all the other cases, we can write $|G|=q^a\cdot\prod_i\Phi_{m_i}^{b_i}(q)$ for some power $q=r^f$ of r, $a,b_i>0$, and $m_i>0$ pairwise distinct, where $\Phi_m(t)$ is the mth cyclotomic polynomial in t. According to [9, §4.10.2], if p divides exactly one $\Phi_{m_i}(q)$, then the Sylow p-subgroups of G are abelian (in fact they are homocyclic of rank b_i). We will now assume that p divides $\Phi_{m_i}(q)$ for more than one m_i.

Here we consider the case $G=S=G_2(q)$. The above condition on p implies that there is some $\epsilon=\pm1$ such that $p=3|(q-\epsilon)$. Then, by [18, Table 5.1], $S>X\cong SL_3^\epsilon(q)$. Now if $9|(q-\epsilon)$, then, as noted above, X and S contain an element of order 9. More generally, according to [20], S contains an element a of order 3, with centralizer of type $A_1(q)\cdot C_{q-\epsilon}$, whence $|\mathbf{C}_S(a)|_3=(q-\epsilon)_3^2$ and $3||a^S|$.

Similarly, if $G=S={}^3D_4(q)$, then again $p=3|(q-\epsilon)$ for some $\epsilon=\pm1$. Then, according to [20], S contains an element a of order 3, with centralizer of type $A_1(q^3)\cdot C_{q-\epsilon}$, whence $|\mathbf{C}_S(a)|_3=3(q-\epsilon)_3^2$ and $3||a^S|$.

Assume that $p=3$ and S is one of the remaining exceptional groups. Then again $p|(q-\epsilon)$ for some $\epsilon=\pm1$, and S contains a subquotient $X\cong\Omega_9(q)$. As mentioned in the proof of Lemma 2.5, then X contains an element of order 9, a contradiction.

(iii) It remains to consider the case $p=5$ and $p|\Phi_{m_i}(q)$ for $i=1,2$ and $m_1\neq m_2$. This happens precisely when

(a) $5|(q-\epsilon)$ for some $\epsilon=\pm1$, and $G\in\{E_6^\pm(q),E_7(q),E_8(q)\}$, or

(b) $5|(q^2+1)$ and $G=E_8(q)$.

In the case of (a), note that G contains a subgroup $X\cong SL_6^\epsilon(q)$, according to [18, Table 5.1]. In turn, X contains $Y\cong GL_5^\epsilon(q)>C_{q^5-\epsilon}$ and so it contains an element of order 25, a contradiction.

Suppose we are in the case of (b). According to [18, Table 5.1], $G = S$ contains a maximal subgroup $X \cong SU_5(q^2) \cdot C_4$. Now if $25|(q^2+1)$, then X and G contain an element of order 25, a contradiction. In general, we can find an element $z \in \mathbf{Z}(X')$ of order 5. We claim that $C := \mathbf{C}_G(z) \leq X$. (Indeed, $Z := \mathbf{Z}(X') = \langle z \rangle$ is normalized by X which is maximal in G, whence $\mathbf{N}_G(Z) = X$ or $Z \lhd G$. In the latter case, since G is perfect, we would have that $Z \leq \mathbf{Z}(G) = 1$, a contradiction. Hence $X = \mathbf{N}_G(Z) \geq \mathbf{C}_G(z)$.) On the other hand, also according to [18, Table 5.1], G contains a subgroup Y of type $C_d^2 \cdot (P\Omega_8^+(q))^2$, where $d = \gcd(2, q-1)$. Choosing an element of order 5 lying in one factor $P\Omega_8^+(q)$, we see that G contains an element $y \in Y$ of order 5 such that $\mathbf{C}_G(y)$ contains a subquotient $\cong P\Omega_8^+(q)$. In particular, $|\mathbf{C}_G(y)|$ is divisible by q^3-1. But obviously $|X|$ is not divisible by q^3-1. It follows that $|y^G| \neq |z^G|$, a contradiction. □

Proof of Theorem 2.1. As mentioned above, by the main result of [23], we may assume that $p \in \{3, 5\}$. Suppose that G satisfies the hypothesis of Theorem 2.1 and that $P \in \mathrm{Syl}_p(G)$ is non-abelian.

Step 1. We have that $\mathbf{O}_p(G) = 1$ and may also assume that $\mathbf{O}_{p'}(G) = 1$.

Indeed, suppose $Q := \mathbf{O}_p(G) \neq 1$. Then $1 \neq \mathbf{Z}(Q) \leq P$. By the assumption, every nontrivial cyclic p-subgroup of P is conjugate to some subgroup of $\mathbf{Z}(Q) \lhd G$, whence $\mathbf{Z}(Q) = P$ and P is abelian. So we have that $\mathbf{O}_p(G) = 1$. Next, by Sylow's Theorem, two non-cyclic p-subgroups of G are conjugate in G if and only if their images are conjugate in $G/\mathbf{O}_{p'}(G)$. Furthermore, the Sylow p-subgroups of $G/\mathbf{O}_{p'}(G)$ are isomorphic to P, so they are not abelian. Replacing G by $G/\mathbf{O}_{p'}(G)$, we may assume that $\mathbf{O}_{p'}(G) = 1$.

Step 2. We have that G has a unique minimal normal subgroup K, which is non-abelian of order divisible by p.

Suppose that K_i are two distinct minimal normal subgroups of G, where $i = 1, 2$. Hence $1 = K_1 \cap K_2$. Now, by Step 1 we know that K_i contains nontrivial cyclic p-subgroups Q_i for $i = 1, 2$ as $\mathbf{O}_{p'}(G) = 1$, and Q_1 and Q_2 are not G-conjugate. Hence, G has a unique minimal normal subgroup K of order divisible by p. Now if K is abelian, then $\mathbf{O}_p(G) \neq 1$, contrary to Step 1.

Step 3. We have that K is simple non-abelian and $K = \mathbf{O}^{p'}(G)$. Furthermore, K satisfies the assumptions of Theorem 2.2.

Since K is minimal normal in G, we see that $K = S_1 \times \cdots \times S_n \cong S^n$, where the S_i's are non-abelian simple groups of order divisible by p which are transitively permuted by G.

Suppose that $n > 1$. Then we can find a p-element $1 \neq x \in S$ and consider the p-elements

$$y = (x, 1, \ldots, 1),\ z = (x, x, 1, \ldots, 1)$$

in K. Now the cyclic p-subgroups $\langle y \rangle$ and $\langle z \rangle$ are not G-conjugate, a contradiction.

Thus K is simple non-abelian. The uniqueness of K implies that $\mathbf{C}_G(K) = 1$ and so G embeds in $\mathrm{Aut}(K)$. Since all cyclic p-subgroups of G are conjugate to a subgroup of K, we see that $K \geq \mathbf{O}^{p'}(G)$, whence $K = \mathbf{O}^{p'}(G)$ by the minimality of K.

It is obvious that $\exp(P) = p$. Moreover, for any two nontrivial p-elements in K, their centralizers in G are G-conjugate. Hence also their centralizers in K are G-conjugate; in particular, these K-centralizers have the same order.

Step 5. Now we apply Theorem 2.2 to $K = \mathbf{O}^{p'}(G)$. Note that in the case $p = 5$ and $K \cong Th$, $G = K$ since $G \hookrightarrow \mathrm{Aut}(K) \cong K$. Also, the Sylow p-subgroups of K are isomorphic to P and so are non-abelian. Now we see that G satisfies the conclusions (ii) or (iii) of Theorem 2.1 and so we are done. □

3. Related Questions on Blocks and Fusion Systems

In this section we ask some related questions and prove a few results. If B is a p-block of G, recall that a B-subsection (u, b_u) is called *major* if the defect groups of b_u are also defect groups of B. Inspired by the main result of [23], we propose the following.

Question 3.1. *Let B be a p-block of a finite group, and suppose that all B-subsections are major. What can be said about the structure of the defect groups of B?*

The main result of [23] provides an answer to Question 3.1 whenever $3 \neq p \neq 5$ and B is the principal block: the defect groups of B in this case should be abelian. Another partial answer is given by the following result which we prove below.

Theorem 3.2. *Let B be a p-block of a finite p-solvable group G. Then all B-subsections are major if and only if the defect groups of B are abelian.*

Theorem 3.2 generalizes several results in the literature, such as Lemma 4.1 in [21] and parts of the main theorem of [28]. J. Olsson [25] already observed that Theorem 3.2 does not hold for arbitrary groups G.

It is well-known that the fusion system of a p-solvable group on its Sylow p-subgroup is p-solvable (cf. p. 2446 of [5]) and that saturated subsystems of p-solvable fusion systems are again p-solvable (cf. p. 2431 of [5]). Thus the fusion system of a p-block of a p-solvable group on its defect group is p-solvable and therefore constrained. (For background on fusion systems we refer to [1] and [6].) Thus Theorem 3.2 is a consequence of the following result on fusion systems. We recall that elements $x, y \in P$ are called *$\mathcal{F}$-conjugate* if $f(x) = y$ for a morphism $f : \langle x \rangle \longrightarrow \langle y \rangle$ in $\mathcal{F}$.

Theorem 3.3. *Let $\mathcal{F}$ be a constrained fusion system on a finite p-group P. Suppose that every element in P is $\mathcal{F}$-conjugate to an element in $\mathbf{Z}(P)$. Then P is abelian.*

Proof. By Theorem 2.5 in [5], there exists a finite group G with Sylow p-subgroup P such that $\mathbf{O}_{p'}(G) = 1$, $\mathbf{C}_G(\mathbf{O}_p(G)) \subseteq \mathbf{O}_p(G)$ and $\mathcal{F} = \mathcal{F}_P(G)$. Let $Q := \mathbf{O}_p(G)$ and $x \in P$. Then our hypothesis implies that there exists an element $g \in G$ such that

$$gxg^{-1} \in \mathbf{Z}(P) \leq \mathbf{C}_G(Q) \leq \mathbf{Z}(Q)$$

which is normal in G, and therefore $x \in \mathbf{Z}(Q)$. This shows that $P = \mathbf{Z}(Q)$; in particular, P is abelian. □

Next we ask whether the following generalization of Theorem A holds:

Question 3.4. *Let B be a p-block of a finite group G, and suppose that all nontrivial B-subsections are conjugate in G. Are the defect groups of B necessarily either elementary abelian or extraspecial of order p^3 and exponent p?*

Since again Question 3.4 is mainly concerned with the fusion system of B we may formulate an even more general question for fusion systems:

Question 3.5. *Let $\mathcal{F}$ be a saturated fusion system on a finite p-group P, and suppose that any two nontrivial cyclic subgroups of P are $\mathcal{F}$-conjugate. Is P necessarily abelian or extraspecial of order p^3 and exponent p?*

If P is extraspecial of order p^3 and exponent p then the saturated fusion systems on P are described in [27]. It follows from these results that $p \in \{3, 5, 7\}$ whenever any two nontrivial elements in P are $\mathcal{F}$-conjugate. Also, if B is a p-block of a finite group with extraspecial defect groups of order p^3 and exponent p such that any two nontrivial B-subsections are conjugate, then the results in [27] and [14] imply that $p \in \{3, 5\}$.

There are several ways to measure the complexity of a block, including to study the structure of a defect group D of B, or the quantities $k(B)$ or $l(B)$. (As usual, here $k(B) = |\mathrm{Irr}(B)|$ and $l(B) = |\mathrm{IBr}(B)|$ where $\mathrm{Irr}(B)$ denotes the set of irreducible ordinary characters of G associated to B, and $\mathrm{IBr}(B)$ denotes the set of irreducible Brauer characters of G associated to B.) Another interesting invariant of the block B is the difference $k(B) - l(B)$. It was already known by Richard Brauer that $k(B) - l(B) = 0$ if and only if $D = 1$ (if and only if $k(B) = l(B) = 1$). From this point of view, the next natural step seems to be to study blocks with

$$k(B) - l(B) = 1\,,$$

which is one of the problems that led us to the main result of this paper. An important formula by Brauer shows that every p-block B of a finite group G satisfies

$$k(B) = \sum_{(u,b_u)} l(b_u)$$

where (u, b_u) ranges over a transversal for the conjugacy classes of B-subsections.

In particular, Brauer's formula implies that, for a p-block B with $k(B) - l(B) = 1$, all nontrivial B-subsections are conjugate. In this case, a result by Fujii [7] shows that the Cartan matrix of B has determinant $|D|$. Moreover, the stable center $\overline{Z}(B) := Z(B)/Z^{\mathrm{pr}}(B)$ is a symmetric algebra, by the results in [13]. In fact, the stable center of B has dimension 2. Theorem 1.1 in [13] implies that, if B is the principal block of G and $k(B) - l(B) = 1$ then the Sylow p-subgroups of G are elementary abelian. Next, we prove more generally the following result.

Theorem 3.6. *Let B be a p-block of a finite group G with $k(B) - l(B) = 1$. Suppose that the fusion system of B is nonexotic (for instance, if B is the principal block or if G is p-solvable). Then the defect groups of B are elementary abelian.*

Recall that a fusion system $\mathcal{F}$ on a finite p-group P is called *nonexotic* if there exists a finite group X with Sylow p-subgroup P such that $\mathcal{F}$ is the fusion system $\mathcal{F}_P(X)$ on P coming from X. (Otherwise the fusion system $\mathcal{F}$ is called *exotic*.) It seems to be an open question whether all fusion systems coming from blocks are nonexotic. A negative answer to Question 3.4 would provide an example of an exotic block fusion system.

Proof of Theorem 3.6. Assume that B has a nonabelian defect group D. Since the fusion system $\mathcal{F}$ of B on D is nonexotic, Theorem A implies that $p \in \{3, 5\}$ and that D is extraspecial of order p^3 and exponent p.

Suppose first that $p = 5$. Then $\mathcal{F}$ is the fusion system of the sporadic simple Thompson group Th on its Sylow 5-subgroup. Now Proposition 6.1 in [11] shows that B is Morita equivalent to the principal 5-block B_0 of Th. In particular, we have $k(B_0) - l(B_0) = 1$. Let (u, b_u) be a nontrivial B_0-subsection. Then b_u is the principal 5-block of $\mathbf{C}_{Th}(u)$, and $l(b_u) = 1$. Thus $\mathbf{C}_{Th}(u)$ is 5-nilpotent, a contradiction.

It remains to consider the case $p = 3$. Let (u, b_u) denote a nontrivial B-subsection, and denote by $\overline{b}_u$ the unique 3-block of $\mathbf{C}_G(u)/\langle u \rangle$ dominated by b_u. Then $1 = l(b_u) = l(\overline{b}_u)$, and $\overline{b}_u$ has an elementary abelian defect group of order 9. By Theorem A, we may assume that $\mathcal{F}$ is the fusion system of ${}^2F_4(2)'$ or J_4 on its Sylow 3-subgroup. Thus the inertial quotient of B is isomorphic to D_8 or SD_{16} respectively, by the results in [27] (cf. [22]). It follows easily that the inertial quotient of $\overline{b}_u$ is isomorphic to C_4 or Q_8 respectively. However, if the inertial quotient of $\overline{b}_u$ is isomorphic to C_4, then the results of [15] lead to the contradiction $l(\overline{b}_u) = 4$.

Thus we may assume that the inertial quotient of $\overline{b}_u$ is isomorphic to Q_8. Then the arguments in [15] show that there are only two $\overline{b}_u$-subsections, and we obtain $k(\overline{b}_u) = 2$. However, then the defect groups of $\overline{b}_u$ have order 2, a contradiction. □

Examples of p-blocks B with $k(B) - l(B) = 1$ are the p-blocks of multiplicity one introduced in [21] by G. Michler. We recall that the *multiplicity* $\mu(B)$ of a p-block B

is defined by

$$\mu(B) := \max\{c_{ii} - 1 : i = 1, \ldots, l(B)\}$$

where $C = (c_{ij})$ denotes the Cartan matrix of B. Thus $\mu(B) = 1$ if and only if all diagonal entries in the Cartan matrix of B are equal to 2.

In [21], Michler showed that a block B of multiplicity one satisfies $k(B) - l(B) = 1$, and that all irreducible ordinary characters associated to B have height zero. Thus, Brauer's Height Zero Conjecture predicts that B should have abelian defect groups. In fact, in [21], it is shown that a p-block B of a finite group G with multiplicity one has abelian defect groups if G is p-solvable or if $p = 2$.

The following more general result is now a consequence of Theorem 3.6:

Corollary 3.7. *Let B be a p-block of a finite group G with multiplicity one. Suppose that the fusion system $\mathcal{F}$ of B is nonexotic (e.g. B is the principal block or G is p-solvable). Then the defect groups of B are elementary abelian.*

4. Blocks with few characters

To finish this paper, we now concentrate on the subject of blocks with few characters. In this context, blocks B with $k(B) - l(B) = 1$ appear quite naturally, as we shall see. First of all, let us summarize a few facts on a p-block B of a finite group G with defect group D having a small number of characters. We have the following:

(i) We have $k(B) = 1$ if and only if $D = 1$.
(ii) We have $k(B) = 2$ if and only if $|D| = 2$ (see [2]).
(iii) If $k(B) \leq 4$ and $l(B) = 1$, then $|D| = k(B)$ (see [17]).
(iv) If $k(B) = 5$ and $l(B) = 1$, then $D \in \{C_5, D_8, Q_8\}$ (see [3]).

In this paper we conjecture that if $k(B) = 3$ then $|D| = 3$, and we prove that this is a consequence of the Alperin-McKay conjecture. First, we take care of the case where the defect group is normal.

Theorem 4.1. *Let B be a p-block of a finite group G with normal defect group D, and suppose that $k(B) = 3$. Then $|D| = 3$.*

Proof. By results of Fong and Reynolds, we may assume that D is a Sylow p-subgroup of G, and that $Z := \mathbf{O}_{p'}(G)$ is cyclic and central in G. By [17], we may also assume that $l(B) = 2$. Then G acts transitively on $D \setminus \{1\}$ by conjugation; in particular, D is elementary abelian. We write $|D| = p^d$. By the Hall-Higman Lemma, the kernel of the action of G on D is ZD. By a result of Passman [26], apart from finitely many exceptions, G/ZD is isomorphic to a subgroup of the semilinear group $T(p^d)$. Here $T(p^d)$ denotes the semidirect product of the multiplicative group $\mathbb{F}_{p^d}^\times$ of the finite field $\mathbb{F}_{p^d}$ with the Galois group Γ of $\mathbb{F}_{p^d}$ over $\mathbb{F}_p$. In particular, G/ZD has a cyclic normal subgroup H/ZD whose order s divides $p^d - 1$ such that G/H is

cyclic of order t dividing d. Since G/ZD acts transitively on $D \setminus \{1\}$ we also have $(p^d - 1) \mid |G : ZD| = st$.

It is well-known that $\mathrm{IBr}(B) = \mathrm{IBr}(G|\zeta)$ for some $\zeta \in \mathrm{IBr}(Z)$. Let us consider $\mathrm{IBr}(H|\zeta)$. On the one hand, $|\mathrm{IBr}(G|\zeta)| = |\mathrm{IBr}(B)| = l(B) = 2$ implies that G has at most two orbits on $\mathrm{IBr}(H|\zeta)$. Moreover, each of these orbits has length at most $|G : H| = t$. Thus $|\mathrm{IBr}(H|\zeta)| \leq 2t \leq 2d$.

On the other hand, we have $ZD/D \leq \mathbf{Z}(H/D)$. Since H/ZD is cyclic, H/D has to be abelian; in particular, we have $|\mathrm{IBr}(H|\zeta)| = |H : ZD| = s$. Thus $s = |\mathrm{IBr}(H|\zeta)| \leq 2d$, and $p^d - 1 \leq |G : ZD| \leq st \leq 2d^2$.

If $p = 2$, then our result follows easily since $k_0(B) \equiv 0 \pmod 4$ for $d \geq 2$. Thus we may assume that $p \geq 3$.

If $d = 1$ then our result follows easily from the Brauer-Dade theory of blocks with cyclic defect groups. Thus we may assume that $d \geq 2$ and $p \geq 3$.

If $d = 2$ then $p^2 \leq 1 + 8 = 9$, i.e. $p = 3$. This case leads to a contradiction by making use of the results in [15]. Thus we may assume that $d \geq 3$ and $p \geq 3$, so that $3^d \leq p^d \leq 1 + d^2$; however, this is impossible.

It remains to deal with the exceptional cases in Passman's Theorem; so we may assume that

$$|D| \in \{3^2,\ 5^2,\ 7^2,\ 11^2,\ 19^2,\ 23^2,\ 29^2,\ 59^2,\ 3^4\}.$$

Suppose first that $d = 2$, and choose a nontrivial B-subsection (u, b_u). Then b_u dominates a unique block $\overline{b}_u$ of $\mathbf{C}_G(u)/\langle u \rangle$, and $\overline{b}_u$ has defect 1. Since $1 = l(b_u) = l(\overline{b}_u)$ we conclude that $\overline{b}_u$ has inertial index 1. Thus b_u has inertial index 1 as well, and G/ZD acts regularly on $D \setminus \{1\}$. Hence G/Z is a Frobenius group with Frobenius kernel ZD/Z and Frobenius complement G/ZD. In particular, the Sylow subgroups of G/ZD are cyclic or (generalized) quaternion. Thus the Schur multiplier of G/ZD is trivial. Hence we may assume that $Z = 1$. But then B is the only p-block of G, so that G has class number 3. This implies that $|G| \leq 6$, a contradiction.

We are left with the case $|D| = 3^4$. In this case G/Z is a doubly transitive permutation group of degree 3^4. It is well-known that $|G/Z| = 2^k 3^4 5$ with $k \in \{5, 6, 7\}$ (see Example XII.7.4 in [12]). Using GAP [8], it is easy to work out the structure of G/Z and to derive a contradiction. □

Theorem 4.2. *Let B be a p-block of a finite group G with defect group D such that $k(B) = 3$. Moreover, suppose that $k_0(B) = k_0(b)$ where b denotes the Brauer correspondent of B in $N_G(D)$. Then $|D| = 3$.*

Here $k_0(B)$ denotes the number of irreducible ordinary characters of height zero associated to B. Thus the conclusion of this theorem is satisfied if the Alperin-McKay Conjecture holds.

Proof of Theorem 4.2. It is well-known that b has defect group D. Then b dominates a unique p-block $\overline{b}$ of $N_G(D)/\Phi(D)$, and $\overline{b}$ has defect group $\overline{D} := D/\Phi(D)$ which is

abelian and normal in $N_G(D)/\Phi(D)$. Moreover, we have

$$k(\bar{b}) = k_0(\bar{b}) \leq k_0(b) = k_0(B) \leq k(B) = 3.$$

If we assume that $k(\bar{b}) \leq 2$, then we get $|\overline{D}| \leq 2$. Thus D is a cyclic 2-group which is impossible. This shows that we must have $k(\bar{b}) = 3$.

Since $\overline{D}$ is normal in $N_G(D)/\Phi(D)$, Theorem 4.1 implies that $|\overline{D}| = 3$. Thus D is cyclic, and the Brauer-Dade theory yields the result. □

It is known that $k_0(B) \geq 2$ for every block B with positive defect (see [24]). Hence, one may ask what can be said in case $k_0(B) = 2$. A similar proof as above shows that the Alperin-McKay Conjecture implies the following generalization of Brandt's result:

$$k_0(B) = 2 \Longleftrightarrow k(B) = 2 \Longleftrightarrow |D| = 2.$$

Finally, in this context of blocks with few characters, it might be of interest to recall that Héthelyi and Külshammer conjectured in [10] that

$$k(B) \geq 2\sqrt{p-1}$$

for arbitrary p-blocks B of arbitrary finite groups.

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